USER GUIDE · CAPABILITIES AT A GLANCE · v3.1.0

MS Word Parser

A digital-forensics tool that pulls the hidden evidence out of Microsoft Word files — authorship, editing sessions, comments, tracked changes, protection and the ZIP container itself — and lays it out in Excel or SQLite for analysis.

`.docx` `.dotx` `.dotm` `.docm`

GET STARTED

Install with the GUI (Python 3.9+)

```
python3 -m pip install "ms-word-parser[gui]"
```

Launch the GUI

```
parse-docx-gui
```

(`parse-docx -g` also works)

See every command-line option

```
parse-docx -h
```

01 How it works

INPUT

Choose your documents

Select individual files

Point at a folder — optionally recurse through every subfolder

Ingest a text file listing full paths, one per line

→

PROCESS

Parse in depth

Triage or Full mode, plus optional hashing and timeline

Runs files in parallel across all CPU cores

Recovers parts from damaged archives; detects encrypted and rights-managed files

→

OUTPUT

Review the results

Excel workbook (.xlsx), with an analysis-tips sheet

SQLite database (.db), with indexes and ready-made views

A log file of everything that happened

`ms-word-parser-YYYYMMDD_HHMMSS.*`

02 Two processing modes

START HERE

Triage

A fast first pass across thousands of documents. Lighter output and a quicker run, so you can spot what matters.

PRODUCES

Document Summary · Metadata · Comments · Excel Tips

DEEP DIVE

Full

Everything in Triage, plus the heavy artifacts. Best run on files flagged during triage, or on large sets with SQLite output.

ADDS

RSIDs · Archive Files · Track Changes · Extended Comments · Comment Reactions · Comments IDs · People · Document Protection · Range Permissions · Rights Management · Content Types · Headers & Footers · Footnotes & Endnotes · Custom Properties · Item XML · Ink XML

#

Add-on: Hash files

MD5 of each document and of every part inside it — to attest integrity later and to find duplicates.

Add-on: Timeline

Merges created, modified, printed, comment, reaction, archive and ink timestamps into one chronological view — with a visual timeline chart in Excel.

03 What it reveals

Authorship & history

Who wrote it, and when?

Author and last modified by

Created, modified and last-printed dates

Revision count and total editing time

Template, application and version

Editing sessions (RSIDs)

Which documents are related?

RSID Root links files derived from the same original document

rsidR values tie text to the save session it was typed in

Shared unused rsidR values hint at the same Windows user

Collaboration

Who else touched it?

Comments and replies, with resolved status

Comment reactions and the people involved

Tracked insertions and deletions

Ink annotations with timestamps

Protection & security

Was access restricted?

Encrypted-file detection

Rights Management (IRM): issued date and validity window

Editing restrictions and range-permission editors

The ZIP container

What does the package itself show?

Every part inside the .docx, with size and timestamp

ZIP attributes and extra-field bytes per part

Flags files likely created by software other than Word

Document structure

What is inside the body?

Paragraph, run, text and table-row tag counts

Document IDs (w14 / w15 / w16)

Hyperlinks, headers, footers, footnotes and endnotes

Spelling and grammar check state

04 Recommended workflow

1

Triage everything

Run Triage across the whole collection.

2

Filter

Sort and filter the summary and metadata to find documents of interest.

3

Make a list

Copy their full paths into a text file, one per line.

4

Go deep

Re-run in Full mode using Ingest and a compiled list of file paths. For large datasets, choose SQLite output.

PROVEN AT SCALE

15K+

files parsed in Full mode in a single run

3m 30s

total processing time for that run

14M+

RSID records held in one SQLite table

Processing time varies with the capabilities of the computer running it.

COMPANION QUERIES · ON GITHUB

27 ready-made SQLite queries

Files that share a specific rsidR value

Last modified by someone other than the author

Low editing time but more than one page

Pending tracked changes with tracking turned off

Encrypted and IRM-protected files

Download them from the `sqlite_queries` folder of the GitHub repository; they are not installed with the tool. The database also includes views for Aggregated Comments and the Timeline.

TWO WAYS TO RUN

Graphical User Interface (GUI)

Launch with `parse-docx-gui`. Live progress (total, errors, remaining), plus Stop, Reset and one-click access to the log and output folder.

Command line

Every option as a flag, ideal for scripting. Flags without values can be combined, as in `-fresHT`. Press Space for a status update, or Q to quit.

ⓘ **Good to know:** page, line and character counts come from the document's own metadata and can be out of date. Re-saving a file to correct them changes its modified date and author — always work on copies.

MS Word Parser · by Jacques Boucher and Corey Forman

github.com/jjrboucher/MS-Word-Parser